Mohammed Abdul Omer

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Professional Summary

AI/ML Engineer specializing in production-grade Generative AI systems, RAG pipelines, and multimodal AI development. Experienced in architecting end-to-end LLM-powered applications using LangChain, FastAPI, and vector databases — deployed into live workflows serving real users. Currently building Emotion Echo, a real-time multimodal AI system combining Computer Vision, Speech Processing, and Agentic AI.

Technical Skills

Languages: Python, SQL

AI / ML Frameworks: PyTorch, Scikit-learn, Hugging Face Transformers, OpenCV, LangChain, LlamaIndex

Generative AI: RAG Pipelines, Vector Databases, Prompt Engineering, LLM Fine-tuning, Agentic AI, OpenAI API, Groq API, LLaMA 3

Backend & APIs: FastAPI, Django, REST APIs

Tools & Platforms: Git, GitHub, Docker, Vercel, Hugging Face Spaces, Jupyter, VS Code

Experience

Artificial Intelligence Engineer Intern — March 2025 – June 2025
TechZone Software Academy for Training and Research — *Hyderabad, India*

- Architected and deployed **4 production-grade AI systems** for a live ed-tech platform serving students and educators across Hyderabad, reducing manual academic workload across departments.
- Engineered a **RAG-powered AI tutoring system** using LangChain and vector databases, delivering contextual real-time academic support and significantly reducing educator response overhead.
- Designed and shipped an **automated PTE mock-test engine** with LLM-based scoring across reading, writing, and listening modules — improving student test readiness through personalized AI feedback.
- Built an **MCQ generation and attendance automation system**, eliminating manual tracking workflows and streamlining academic operations across multiple departments.
- Delivered all systems as **production-ready FastAPI services** with REST-based orchestration, achieving continuous reliability through prompt optimization.

Artificial Intelligence Intern — November 2024
RAM Innovative Infotech (College Training Program) — *Hyderabad, India*

- Developed a **disease prediction ML model** using Python and Scikit-learn applying supervised learning classification on medical datasets, organized in collaboration with Lords Institute CSE-AIML.
- Gained hands-on exposure to Django for web-based AI integration and end-to-end data preprocessing pipelines.

Projects

Building Safety Smoke Detection | *Python, TensorFlow, Scikit-learn, YOLOv8, OpenCV, Django* — 2025 – 2026

- **Final Year Major Project** — Engineered a dual-module intelligent fire and smoke detection system trained on a real-world IoT sensor dataset of **62,630 readings** across 13 sensor channels.
- Trained **7 ML classifiers** (Random Forest, SVM, Gradient Boosting, KNN, and others) achieving **AUC-ROC scores above 0.999**; deployed a **MobileNetV2 CNN** via transfer learning reaching **96.98% validation accuracy** on fire/no-fire image classification.
- Integrated **YOLOv8 object detection** to draw real-time bounding boxes around fire and smoke regions; delivered the full system as a role-based Django web application deployed on Railway.

LLM-Powered Doubt Tutor | *LangChain, Ollama, Streamlit, Python* — 2025

- Built a local AI doubt-solving assistant using LangChain and Ollama (local LLMs) supporting PDF, image, and text uploads — delivering contextual responses with clean Streamlit UI and chat export.

Education

Lords Institute of Engineering & Technology — January 2022 – May 2026
Bachelor of Engineering, Computer Science Engineering (AI & ML) — *Hyderabad, India*

- Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Structures & Algorithms, Database Management Systems